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Problem 1: Variational Auto-Encoder (VAE)

Original AI-Generated Summary

"The Variational Auto-Encoder (VAE) introduced the SGVB (Stochastic Gradient Variational Bayes) estimator to perform efficient inference in directed graphical models with continuous latent variables. The architecture consists of an Encoder network that maps the input x directly to a single point vector z in a lower-dimensional latent space. To ensure the model remains end-to-end differentiable, the authors introduce the Reparameterization Trick, which involves adding uniform noise (sampled from $U[-1, 1]$) to the latent vector before it is passed to the decoder. The model is optimized by maximizing the Evidence Lower Bound (ELBO). This objective function consists of a discriminative term used to predict the class labels of the data, and a KL Divergence term that acts as a penalty to ensure the approximate posterior $q_\phi(z|x)$ is as dissimilar as possible from the prior $p(z)$. Finally, the paper demonstrates through experiments on the Frey Faces dataset that for complex image data, the prior $p(z)$ must be a learned mixture of Gaussians to achieve high-quality reconstruction."

Identified Discrepancies

1. **Incorrect Statement:** "The architecture consists of an Encoder network that maps the input x directly to a single point vector z in a lower-dimensional latent space."

Technical Correction: The encoder does not output a deterministic, single-point latent vector. Instead, it generates the parameters (typically mean and variance) for a probability distribution $q_\phi(z|x)$ over the latent space, and the latent vector z is then stochastically sampled from this distribution.

Supporting Evidence from Paper: In Section 3 (Example: Variational Auto-Encoder), the authors state: "In this case, we can let the variational approximate posterior be a multivariate Gaussian with a diagonal covariance structure . . . where the mean and s.d. of the approximate posterior, $\mu^{(i)}$ and $\sigma^{(i)}$, are outputs of the encoding MLP". This is formally defined in Equation 9:

$$\log q_\phi(z|x^{(i)}) = \log \mathcal{N}(z; \mu^{(i)}, \sigma^{2(i)} I)$$

2. **Incorrect Statement:** "To ensure the model remains end-to-end differentiable, the authors introduce the Reparameterization Trick, which involves adding uniform noise (sampled from $U[-1, 1]$) to the latent vector before it is passed to the decoder."

Technical Correction: The reparameterization trick relies on sampling from a standard normal distribution, not a uniform distribution. This noise is then scaled and shifted using the variance and mean outputted by the encoder.

Supporting Evidence from Paper: In Section 3 (Example: Variational Auto-Encoder), the paper specifies: “As explained in section 2.4, we sample from the posterior $z^{(i,l)} \sim q_\phi(z|x^{(i)})$ using $z^{(i,l)} = g_\phi(x^{(i)}, \epsilon^{(l)}) = \mu^{(i)} + \sigma^{(i)} \odot \epsilon^{(l)}$ where $\epsilon^{(l)} \sim \mathcal{N}(0, I)$ ”.

3. **Incorrect Statement:** “This objective function consists of a discriminative term used to predict the class labels of the data, and a KL Divergence term that acts as a penalty to ensure the approximate posterior $q_\phi(z|x)$ is as dissimilar as possible from the prior $p(z)$.”

Technical Correction: The ELBO relies on a generative expected negative reconstruction error, not a discriminative classification term. Furthermore, the KL divergence term acts to minimize the difference between the approximate posterior and the prior, encouraging similarity rather than dissimilarity.

Supporting Evidence from Paper: In Section 2.3 (The SGVB estimator and AEVB algorithm), the text notes: “The first term is (the KL divergence of the approximate posterior from the prior) acts as a regularizer, while the second term is a an expected negative reconstruction error.” Furthermore, the regularizing effect is described as “encouraging the approximate posterior to be close to the prior $p_\theta(z)$ ”. The components are shown in Section 2.2 (The variational bound), Equation 3:

$$\mathcal{L}(\theta, \phi; x^{(i)}) = -D_{KL}(q_\phi(z|x^{(i)})||p_\theta(z)) + \mathbb{E}_{q_\phi(z|x^{(i)})}[\log p_\theta(x^{(i)}|z)]$$

4. **Incorrect Statement:** “Finally, the paper demonstrates through experiments on the Frey Faces dataset that for complex image data, the prior $p(z)$ must be a learned mixture of Gaussians to achieve high-quality reconstruction.”

Technical Correction: The framework specifically utilizes a fixed, standard isotropic Gaussian distribution for the prior, which explicitly lacks any learnable parameters.

Supporting Evidence from Paper: In Section 3 (Example: Variational Auto-Encoder), the authors explicitly state: “Let the prior over the latent variables be the centered isotropic multivariate Gaussian $p_\theta(z) = \mathcal{N}(z; 0, I)$. Note that in this case, the prior lacks parameters.”

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Problem 2: Black Box Variational Inference (BBVI)

Original AI-Generated Summary

"The Black Box Variational Inference (BBVI) framework was designed to automate variational inference. The authors propose an algorithm that only requires the user to provide the analytic gradient of the ELBO with respect to the latent variables z .

The core of the method is the Log-Derivative Trick. To ensure the algorithm converges, the authors introduce Control Variates. In this paper, control variates are used solely to bias the gradient estimates toward the global optimum, ensuring that the stochastic updates never stray into local minima. These functions must be manually derived for every new model to ensure the gradient variance is exactly zero.

To optimize the parameters, the algorithm uses a fixed, large learning rate. Finally, the paper demonstrates BBVI using Stochastic Volatility models, proving the method works even when the model's log-likelihood $\log p(x, z)$ cannot be evaluated numerically."

Identified Discrepancies

1. **Incorrect Statement:** "The authors propose an algorithm that only requires the user to provide the analytic gradient of the ELBO with respect to the latent variables z ."

Technical Correction: The BBVI algorithm does not require analytic gradients with respect to the latent variables. Instead, it optimizes the variational parameters λ using stochastic optimization, where gradients are estimated via Monte Carlo sampling from the variational distribution.

Supporting Evidence from Paper: In Section 2: Black Box Variational Inference, the authors state: "We will form the derivative of the objective as an expectation with respect to the variational approximation and then sample from the variational approximation to get noisy but unbiased gradients". This is formally defined in Equation 2:

$$\nabla_{\lambda} \mathcal{L} = E_q[\nabla_{\lambda} \log q(z|\lambda)(\log p(x, z) - \log q(z|\lambda))]$$

2. **Incorrect Statement:** "In this paper, control variates are used solely to bias the gradient estimates toward the global optimum, ensuring that the stochastic updates never stray into local minima."

Technical Correction: Control variates are introduced exclusively to reduce the variance of the stochastic gradient estimator, not to bias it. By definition, they maintain the unbiased nature of the estimator and do not inherently guarantee convergence to the global optimum.

Supporting Evidence from Paper: In Section 3.2: Control Variates, the text explains: “variance reduction methods work by replacing the function whose expectation is being approximated by Monte Carlo with another function that has the same expectation but smaller variance ... note that $E_q[\hat{f}(z)] = E_q[f]$ as required.”

3. **Incorrect Statement:** “These functions must be manually derived for every new model to ensure the gradient variance is exactly zero.”

Technical Correction: A core contribution of the BBVI framework is that control variates do not require model-specific manual derivations, preserving the “black box” nature of the algorithm. Additionally, while they significantly reduce the variance of the estimator, they do not reduce it to exactly zero.

Supporting Evidence from Paper: In Section 1: Introduction, the authors explicitly state: “We emphasize that these variance reduction methods preserve our goal of black box inference because they do not require computations specific to the model.” Furthermore, Section 3.2: Control Variates details that they “choose a control variate that only depends on the variational distribution”.

4. **Incorrect Statement:** “To optimize the parameters, the algorithm uses a fixed, large learning rate.”

Technical Correction: The algorithm does not use a fixed, large learning rate. It relies on a decreasing learning rate schedule that satisfies the Robbins-Monro conditions to ensure convergence, and later extends this using AdaGrad for adaptive, per-component step sizes.

Supporting Evidence from Paper: In Section 2: Black Box Variational Inference, it is noted that the stochastic optimization “converges to a maximum of $f(x)$ when the learning rate schedule follows the Robbins-Monro conditions”:

$$\sum_{t=1}^{\infty} \rho_t = \infty, \quad \sum_{t=1}^{\infty} \rho_t^2 < \infty$$

Additionally, Section 4.1: AdaGrad introduces adaptive learning rates to address varying scales and noise.

5. **Incorrect Statement:** “... proving the method works even when the model’s log-likelihood $\log p(x, z)$ cannot be evaluated numerically.”

Technical Correction: The BBVI algorithm strictly requires the practitioner to be able to numerically evaluate the joint log-likelihood of the model for a given sample. Without this, the Monte Carlo estimate of the gradient cannot be computed.

Supporting Evidence from Paper: In Section 2: Black Box Variational Inference, the paper states: “From the practitioner’s perspective, this method requires only that he or she write functions to evaluate the model log-likelihood. ... we did not make any assumptions about the form of the model, only that the practitioner can compute the log of the joint $p(x, z_s)$.”

Problem 3: Stochastic Gradient Langevin Dynamics (SGLD)

Original AI-Generated Summary

"The Stochastic Gradient Langevin Dynamics (SGLD) algorithm provides a way to perform Bayesian inference on massive datasets. The authors propose a modification to standard SGD where a fixed amount of constant variance noise is added to the gradient updates at every iteration. This noise ensures that the sampler explores the entire parameter space regardless of the step size.

Crucially, the algorithm eliminates the need for the Metropolis-Hastings (MH) acceptance step by ensuring the step size ϵ_t stays constant throughout training. This allows the chain to maintain a high rejection rate to avoid local minima. The authors prove that as long as the noise is sampled from a Uniform distribution, the distribution of the parameters θ will converge to a single MAP (Maximum A Posteriori) estimate. Finally, the paper demonstrates the method's efficiency using a Logistic Regression model on the MNIST dataset."

Identified Discrepancies

1. **Incorrect Statement:** "Crucially, the algorithm eliminates the need for the Metropolis-Hastings (MH) acceptance step by ensuring the step size ϵ_t stays constant throughout training."

Technical Correction: The summary incorrectly claims that the step size ϵ_t is kept constant. SGLD actually requires annealing the step size (decreasing it towards zero), which causes the discretization error to vanish and the MH rejection rate to approach zero, thereby allowing the MH acceptance step to be safely ignored.

Supporting Evidence from Paper: As stated in Section 3 (Stochastic Gradient Langevin Dynamics): "The proposed update is simply: ... where the step sizes decrease towards zero at rates satisfying (2). This allows averaging out of the stochasticity in the gradients, as well as MH rejection rates that go to zero asymptotically, so that we can simply ignore the MH acceptance steps, which require evaluation of probabilities over the whole dataset, all together."

2. **Incorrect Statement:** "This allows the chain to maintain a high rejection rate to avoid local minima."

Technical Correction: The summary states that the algorithm maintains a high rejection rate but in reality, because the step sizes are annealed to zero, the discretization error vanishes, causing the MH rejection rate to asymptotically approach zero rather than remaining high.

Supporting Evidence from Paper: Again, from Section 3: "This allows averaging out of the stochasticity in the gradients, as well as MH rejection rates that go to zero

asymptotically, so that we can simply ignore the MH acceptance steps, which require evaluation of probabilities over the whole dataset, all together.”

3. **Incorrect Statement:** “The authors prove that as long as the noise is sampled from a Uniform distribution, the distribution of the parameters will converge to a single MAP (Maximum A Posteriori) estimate.”

Technical Correction: This sentence contains two critical errors. First, the injected noise is sampled from a Gaussian (Normal) distribution, not a Uniform distribution. Second, the fundamental purpose of the SGLD algorithm is to transition away from finding a single MAP estimate and instead allow the parameter sequence to converge to the full posterior distribution to capture uncertainty.

Supporting Evidence from Paper: Regarding the noise distribution, the paper explicitly defines it as Normal in Equation 4 from Section 3:

$$\Delta\theta_t = \frac{\epsilon_t}{2} \left(\nabla \log p(\theta_t) + \frac{N}{n} \sum_{i=1}^n \nabla \log p(x_{ti}|\theta_t) \right) + \eta_t$$

$$\eta_t \sim N(0, \epsilon_t) \quad (4)$$

About convergence, in the same section the paper states: “Since this is in turn dominated by the total injected noise which is $O(\sqrt{\epsilon_0})$, this means that the sequence $\theta_{t_1}, \theta_{t_2}, \dots$ will approach a sequence generated by Langevin dynamics with a fixed step size ϵ_0 , so it will converge to the posterior distribution.”

4. **Incorrect Statement:** “Finally, the paper demonstrates the method’s efficiency using a Logistic Regression model on the MNIST dataset.”

Technical Correction: The authors actually evaluated their Logistic Regression model using the a9a dataset (derived from the UCI adult dataset) and not the MNIST image dataset.

Supporting Evidence from Paper: “We applied our stochastic gradient Langevin algorithm to a Bayesian logistic regression model. ... We applied our inference algorithm to the a9a dataset derived by (Lin et al., 2008) from the UCI adult dataset.” (Secion 5.2 Logistic Regression)

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Problem 4: Deep Neural Networks as GPs (NNGP)

Original AI-Generated Summary

"The authors demonstrate that deep neural networks (DNNs) with a finite number of hidden units per layer are equivalent to Gaussian Processes. This equivalence holds because as you add more layers to a network, the Central Limit Theorem (CLT) causes the output to become normally distributed, regardless of the width of the hidden layers.

The paper introduces a recursive method to compute the Neural Tangent Kernel (NTK), which defines the covariance between inputs in a deep GP. They prove that to maintain this GP equivalence, one must use a Sigmoid activation function, as ReLU activations cause the covariance to collapse to zero at every layer. Additionally, the authors find that the GP performance is always strictly worse than the corresponding finite-width trained neural network. Finally, the researchers evaluate their theory using the MNIST dataset, showing that the GP derived from a 10-layer CNN outperforms a standard SVM."

Identified Discrepancies

1. **Incorrect Statement:** "The authors demonstrate that deep neural networks (DNNs) with a finite number of hidden units per layer are equivalent to Gaussian Processes."

Technical Correction: The exact mathematical equivalence between deep neural networks and Gaussian Processes does not hold for finite-width networks; it is strictly established only in the mathematical limit as the width of the hidden layers approaches infinity.

Supporting Evidence from Paper: In Section 1 (Introduction), the authors state: "Consider a deep fully-connected neural network with i.i.d. random parameters. ... in the limit of infinite width the Central Limit Theorem implies that the function computed by the neural network (NN) is a function drawn from a Gaussian process (GP)." This is formalized in Section 2.3 (Gaussian Processes and Deep Neural Networks): "We proceed by taking the hidden layer widths to be infinite in succession ($N_1 \rightarrow \infty$, $N_2 \rightarrow \infty$, etc.)."

2. **Incorrect Statement:** "This equivalence holds because as you add more layers to a network, the Central Limit Theorem (CLT) causes the output to become normally distributed..."

Technical Correction: The application of the Central Limit Theorem is driven by the summation over a large number of independent neuron contributions within a single layer (its width), rather than the sequential addition of layers (its depth).

Supporting Evidence from Paper: In Section 2.2 (Review of Gaussian Processes and Single-Layer Neural Networks), the paper explains: "Because the weight and bias

parameters are taken to be i.i.d., the post-activations ...are independent ... Moreover, since $z_i^1(x)$ is a sum of i.i.d terms, it follows from the Central Limit Theorem that in the limit of infinite width $N_1 \rightarrow \infty$, $z_i^1(x)$ will be Gaussian distributed.”

3. **Incorrect Statement:** “This equivalence holds because as you add more layers to a network, the Central Limit Theorem (CLT) causes the output to become normally distributed, regardless of the width of the hidden layers. ”

Technical Correction: The width of the hidden layers is the fundamental parameter governing this equivalence; the GP correspondence breaks down without taking the limit of the layer width to infinity.

Supporting Evidence from Paper: In Section 2.3 (Gaussian Processes and Deep Neural Networks), the paper notes: “As before, $z_i^l(x)$ is a sum of i.i.d. random terms so that, as $N_l \rightarrow \infty$ any finite collection ...will have joint multivariate Gaussian distribution”. Furthermore, Section 3.1 (Description) observes: “neural network performance more closely resembles NNGP performance with increasing width.”

4. **Incorrect Statement:** “Additionally, the authors find that the GP performance is always strictly worse than the corresponding finite-width trained neural network.”

Technical Correction: Empirical results in the paper demonstrate the opposite; Neural Network Gaussian Processes (NNGPs) serve as highly competitive models that frequently surpass the predictive performance of finite-width neural networks trained via standard optimization methods.

Supporting Evidence from Paper: In Section 1.2 (Summary of Contributions), the authors conclude: “Our experiments reveal that the best NNGP performance is consistently competitive against that of NNs trained with gradient-based techniques, and the best NNGP setting, chosen across hyperparameters, often surpasses that of conventional training”. The Abstract also explicitly notes: “GP predictions typically outperform those of finite-width networks.”

Problem 5: BatchBALD

Original AI-Generated Summary

"BatchBALD is an extension of the BALD objective designed for high-throughput labeling. The authors point out that standard BALD is inefficient for batch acquisition because it underestimates the uncertainty of individual points, leading the model to ignore outliers.

To solve this, BatchBALD maximizes the sum of the individual mutual informations of all points in a candidate batch. This objective ensures that the points in the batch are as similar to each other as possible to provide a 'consensus' update to the model parameters θ . The paper proves that this objective is linearly scalable, meaning that calculating a batch of size 1000 is no more computationally expensive than calculating a batch of size 1.

Furthermore, the authors demonstrate that BatchBALD is particularly effective for Support Vector Machines (SVMs) because they lack the 'overconfidence' found in Neural Networks. Finally, the researchers evaluate their method on the Repeated MNIST dataset, showing that BatchBALD avoids the redundancy that plagues top-K BALD."

Identified Discrepancies

1. **Incorrect Statement:** "The authors point out that standard BALD is inefficient for batch acquisition because it underestimates the uncertainty of individual points, leading the model to ignore outliers."

Technical Correction: Standard BALD does not underestimate individual uncertainty; it evaluates points entirely independently. This independence causes the algorithm to naively select highly similar and redundant samples that share the same information, leading to data inefficiency.

Supporting Evidence from Paper: In Section 1: Introduction, the authors state: "In Gal et al. [13], batch acquisition, i.e. the acquisition of multiple points, takes the top b points with the highest BALD acquisition score. This naive approach leads to acquiring points that are individually very informative, but not necessarily so jointly."

2. **Incorrect Statement:** "To solve this, BatchBALD maximizes the sum of the individual mutual informations of all points in a candidate batch."

Technical Correction: Summing individual mutual informations is the exact mechanism of the standard BALD approach. BatchBALD instead resolves this redundancy by optimizing the joint mutual information between the candidate batch as a whole and the model parameters.

Supporting Evidence from Paper: In Section 3.1: BatchBALD, the paper proposes the solution: “We propose BatchBALD as an extension of BALD whereby we jointly score points by estimating the mutual information between a joint of multiple data points and the model parameters: $a_{BatchBALD}(\{x_1, \dots, x_b\}, p(\omega|\mathcal{D}_{train})) = \mathbb{I}(y_1, \dots, y_b; \omega|x_1, \dots, x_b, \mathcal{D}_{train})$.”

3. **Incorrect Statement:** “This objective ensures that the points in the batch are as similar to each other as possible to provide a ‘consensus’ update to the model parameters θ .”

Technical Correction: The goal of the BatchBALD objective is exactly the opposite: it ensures the selected batch of points is as diverse as possible to provide complementary information and avoid double-counting overlaps.

Supporting Evidence from Paper: In Section 3.1: BatchBALD, the authors note that naive BALD leads to “selecting similar data points instead of diverse ones under maximisation”, whereas BatchBALD “takes overlaps into account... and is more likely to acquire a more diverse cover under maximisation”. Additionally, the caption for Figure 8 explicitly states: “BatchBALD steadily acquires a more diverse set of data points.”

4. **Incorrect Statement:** “The paper proves that this objective is linearly scalable, meaning that calculating a batch of size 1000 is no more computationally expensive than calculating a batch of size 1.”

Technical Correction: The computational cost of BatchBALD is not constant. While the greedy approximation scales linearly with respect to the batch size (avoiding the exponential explosion of exact subset evaluation), computing a large batch is strictly more computationally expensive than computing a batch of size 1.

Supporting Evidence from Paper: In Section 3.4: Efficient implementation, the paper bounds the time complexity of the full algorithm to $\mathcal{O}(bc \cdot \min\{c^b, m\} \cdot |\mathcal{D}_{pool}| \cdot k)$, which clearly grows proportionally with the acquisition size b .

Problem 6: Deep Ensembles

Original AI-Generated Summary

"This paper provides a theoretical justification for deep ensembles by proving that they are equivalent to exact Bayesian inference in neural networks. The authors argue that each model in the ensemble corresponds to an independent sample from the true posterior distribution over weights.

To study this phenomenon, the paper analyzes the loss landscape in weight space and shows that models obtained along a single SGD training trajectory explore multiple distinct modes of the posterior. In contrast, models initialized independently tend to converge to the same mode due to the strong convexity of the loss function.

The authors introduce a diversity-accuracy tradeoff and conclude that increasing diversity always harms predictive accuracy, making large ensembles undesirable. Finally, the experiments demonstrate that subspace sampling methods such as SWAG consistently outperform deep ensembles on out-of-distribution generalization tasks."

Identified Discrepancies

1. **Incorrect Statement:** "This paper provides a theoretical justification for deep ensembles by proving that they are equivalent to exact Bayesian inference in neural networks."

Technical Correction: The paper does not prove an equivalence to exact Bayesian inference. Instead, it highlights a practical gap where deep ensembles empirically outperform theoretically grounded Bayesian neural networks because popular variational Bayesian methods often collapse to a single mode.

Supporting Evidence from Paper: In the Abstract, the authors state: "Bayesian neural networks, which learn distributions over the parameters of the network, are theoretically well-motivated by Bayesian principles, but do not perform as well as deep ensembles in practice, particularly under dataset shift. One possible explanation for this gap between theory and practice is that popular scalable variational Bayesian methods tend to focus on a single mode, whereas deep ensembles tend to explore diverse modes in function space."

2. **Incorrect Statement:** "...models obtained along a single SGD training trajectory explore multiple distinct modes of the posterior. In contrast, models initialized independently tend to converge to the same mode due to the strong convexity of the loss function."

Technical Correction: The study demonstrates the reverse: functions sampled along a single optimization trajectory cluster within the same local mode in function space, whereas independent random initializations successfully explore entirely different, disconnected modes.

Supporting Evidence from Paper: In the Abstract, the authors explicitly note: “Our results show that random initializations explore entirely different modes, while functions along an optimization trajectory or sampled from the subspace of a single mode . . . cluster within a single mode”.

3. **Incorrect Statement:** “The authors introduce a diversity-accuracy tradeoff and conclude that increasing diversity always harms predictive accuracy, making large ensembles undesirable.”

Technical Correction: The diversity-accuracy analysis reveals that independent random initializations are highly desirable precisely because they achieve an optimal balance of being both accurate and diverse, which is a fundamental requirement for effective ensembling.

Supporting Evidence from Paper: In Section 4.3: Diversity versus Accuracy plots, the paper explains: “From a bias-variance trade-off perspective, we require a procedure to produce functions that are accurate (which leads to low bias by aggregation) as well as de-correlated (which leads to lower variance by aggregation).” and observes that “...we observe that random initializations are much more effective at sampling diverse and accurate solutions . . .”

4. **Incorrect Statement:** “Finally, the experiments demonstrate that subspace sampling methods such as SWAG consistently outperform deep ensembles on out-of-distribution generalization tasks.”

Technical Correction: The experimental results confirm that deep ensembles significantly outperform subspace sampling methods under dataset shift because their higher prediction diversity prevents the ensemble from making correlated, overconfident errors on corrupted data.

Supporting Evidence from Paper: In Section 5: Evaluating the Relative Effects of Ensembling versus Subspace Methods, the authors conclude: “...which explains the findings of Ovadia et al. [2019] that deep ensembles outperform other methods under dataset shift. We also observe similar trends when testing on an OOD dataset such as...”

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Problem 7: Automatic Differentiation Variational Inference (ADVI)

Original AI-Generated Summary

"Automatic Differentiation Variational Inference (ADVI) provides a general-purpose framework for variational inference by expressing the Evidence Lower Bound (ELBO) in a transformed parameter space where all latent variables follow a standard normal prior. This transformation ensures that the variational approximation can be optimized without reference to the original model constraints.

The method relies on a reparameterization of the variational distribution that eliminates the need for stochastic gradient estimates, allowing exact gradients of the ELBO to be computed via automatic differentiation. In practice, ADVI is implemented using a mean-field Gaussian approximation, as the paper shows that full-rank covariance structures offer negligible practical benefits.

A key advantage of ADVI is that it avoids the need for model-specific derivations by assuming that all latent variables are independent in the variational family, which simplifies inference without significantly affecting posterior correlations. Finally, the empirical evaluation demonstrates that ADVI consistently matches or exceeds the accuracy of Hamiltonian Monte Carlo across all tested models while being significantly faster."

Identified Discrepancies

1. **Incorrect Statement:** "The method relies on a reparameterization of the variational distribution that eliminates the need for stochastic gradient estimates, allowing exact gradients of the ELBO to be computed via automatic differentiation."

Technical Correction: The summary states that ADVI computes exact gradients and eliminates stochastic estimates but in reality, the ELBO expectation remains intractable, so ADVI relies on Monte Carlo integration to compute noisy, unbiased stochastic gradient estimates, which are then used in a stochastic gradient ascent algorithm.

Supporting Evidence from Paper: "We can now compute the gradients inside the expectation with automatic differentiation. The only thing left is the intractable expectation. MC integration provides a simple approximation: draw samples from the standard Gaussian and evaluate the empirical mean of the gradients within the expectation ... This gives noisy unbiased gradients of the ELBO for any differentiable probability model. We can use these gradients in a stochastic optimization routine to automate variational inference ... Equipped with noisy unbiased gradients of the ELBO, AD implements stochastic gradient ascent (Algorithm 1)." (Section 2.6: Stochastic Optimization)

2. **Incorrect Statement:** "A key advantage of ADVI is that it avoids the need for model-specific derivations by assuming that all latent variables are independent in the variational

family, which simplifies inference without significantly affecting posterior correlations.”

Technical Correction: The summary contradicts fundamental variational inference theory by claiming the independence assumption (mean-field) doesn’t affect posterior correlations. The paper explicitly demonstrates that the mean-field approximation fundamentally fails to capture local correlation structures and systematically distorts the variance.

Supporting Evidences from Paper: “This is because the mean-field approximation ignores off-diagonal terms of the Gaussian covariance. ADVI minimizes the kl divergence from the approximation to the exact posterior; this leads to a systemic underestimation of marginal variances (Bishop, 2006) . . . The mean-field covariance (fig. 7a) fails to capture the locally correlated structure of the full-rank and sampling covariance matrices (figs. 7b and 7c).” (Section 3.1: Accuracy)

3. **Incorrect Statement:** “Automatic Differentiation Variational Inference (ADVI) provides a general-purpose framework for variational inference by expressing the Evidence Lower Bound (ELBO) in a transformed parameter space where all latent variables follow a standard normal prior.”

Technical Correction: The summary confuses the model’s prior distribution with the variational approximation. ADVI does not force the latent variables to follow a standard normal *prior*. Instead, it uses an “elliptical standardization” to reparameterize the *variational approximation* (q) into a standard normal distribution, which allows the gradients to be computed inside the expectation.

Supporting Evidence from Paper: “Consider a transformation S_ϕ that absorbs the variational parameters ϕ ; this converts the Gaussian variational approximation into a standard Gaussian . . . In both cases, the standardization encapsulates the variational parameters; in return it gives a fixed variational density $q(\eta) = Normal(\eta|0, I)$.” (Section 2.5: The Variational Problem in Real Coordinate Space)

4. **Incorrect Statement:** “In practice, ADVI is implemented using a mean-field Gaussian approximation, as the paper shows that full-rank covariance structures offer negligible practical benefits.”

Technical Correction: The summary falsely claims that full-rank covariance structures offer negligible benefits. The paper demonstrates that the mean-field approximation systemically underestimates marginal variances. The authors explicitly recommend the full-rank approximation when accurate posterior variances and local correlation structures are needed, making its practical benefits highly significant.

Supporting Evidence from Paper: “Scientists interested in posterior variances and covariances should use the full-rank approximation. Full-rank ADVI captures posterior correlations, in turn producing more accurate marginal variance estimates.” (Section 3.1: Accuracy - Recommendations)

Problem 8: Laplace Redux - Effortless Bayesian Deep Learning

Original AI-Generated Summary

"This paper revisits the Laplace approximation (LA) as a practical approach to Bayesian deep learning, emphasizing that its effectiveness depends primarily on the choice of curvature approximation used to model posterior uncertainty. In particular, the authors show that different approximations (e.g., diagonal, KFAC, and full-matrix forms) lead to similar predictive performance when applied to modern neural networks.

A key contribution of the paper is demonstrating that applying LA to only the final layer provides a computationally efficient approximation that closely matches full-network Laplace across a wide range of tasks, including out-of-distribution detection. This suggests that most of the useful posterior uncertainty is captured in the last layer of the network.

The authors further argue that marginal likelihood optimization for tuning prior precision is largely unnecessary in practice, as predictive performance is relatively insensitive to this hyperparameter when using well-trained networks.

Finally, the paper concludes that Laplace approximations offer competitive uncertainty estimates compared to variational inference and deep ensembles, particularly when computational cost is taken into account, though their performance is largely invariant to the specific predictive approximation used."

Identified Discrepancies

1. **Incorrect Statement:** "The authors further argue that marginal likelihood optimization for tuning prior precision is largely unnecessary in practice, as predictive performance is relatively insensitive to this hyperparameter when using well-trained networks."

Technical Correction: The summary claims that tuning prior precision via marginal likelihood optimization is unnecessary but actually the paper strongly advocates for it (also known as empirical Bayes), setting it as the default recommended hyperparameter tuning method for their `laplace` library. Furthermore, the authors explicitly demonstrate that tuning the prior precision through marginal likelihood optimization actively improves performance, such as in overcoming catastrophic forgetting.

Supporting Evidence from Paper: In Section 4.4 (Further Applications), the authors state: "In this experiment, we update the LAs after each task as suggested by Ritter et al. [24] and improve upon their result by tuning the prior precision through marginal likelihood optimization during training, following Immer et al. [22]..." Additionally, in Section 3 (Default behavior), they establish this as their standard approach: "...we provide

the following default choices based on our extensive experiments (Section 4)... tune the hyperparameters post-hoc using empirical Bayes.”

2. **Incorrect Statement:** “In particular, the authors show that different approximations (e.g., diagonal, KFAC, and full-matrix forms) lead to similar predictive performance when applied to modern neural networks.”

Technical Correction: The summary incorrectly asserts that the choice of covariance approximation does not significantly impact predictive performance. The paper actually finds different performances based on different approximations.

Supporting Evidence from Paper: In Section 4.1 under the heading *Covariance approximation and structure*, the authors state: “Generally, we find that a more expressive covariance approximation improves performance, as would be expected. ... Diagonal approximations perform significantly worse than KFAC and are therefore not suggested.”

3. **Incorrect Statement:** “A key contribution of the paper is demonstrating that applying LA to only the final layer provides a computationally efficient approximation that closely matches full-network Laplace across a wide range of tasks, including out-of-distribution detection”

Technical Correction: The summary claims that the last-layer Laplace Approximation simply “closely matches” the performance of the full-network LA. In reality, the paper demonstrates that applying the LA post-hoc to only the last layer actually *outperforms* applying it to all weights, because the full-network approach suffers from underfitting.

Supporting Evidence from Paper: In Section 4.1 under the heading *Hyperparameter tuning and parameter inference*, the authors state: “In contrast, applying the LA post-hoc only on the last layer instead of all weights typically yields better performance due to less underfitting, and is significantly cheaper.”

4. **Incorrect Statement:** “Finally, the paper concludes that Laplace approximations offer competitive uncertainty estimates compared to variational inference and deep ensembles, particularly when computational cost is taken into account, though their performance is largely invariant to the specific predictive approximation used.”

Technical Correction: The summary erroneously states that the choice of predictive approximation doesn’t significantly affect performance. The paper explicitly shows the opposite: performance varies noticeably depending on the approximation. Specifically, the authors found that the Laplace bridge performs poorly, while the probit is generally best, and Monte Carlo (MC) can be superior for out-of-distribution tasks.

Supporting Evidence from Paper: In Section 4.1 under the heading *Predictive distribution*, the authors note clear variances in performance: “Considering in- and out-of-distribution (OOD) performance as well as cost, the probit provides the best approximation to the predictive for the last-layer LA. MC integration can sometimes be superior for OOD detection... The Laplace bridge ... typically provides inferior results in our experiments.”